

Assignment 3 - Practice Quiz A

20 questions; 30 minutes suggested. Practice does not affect your grade.

1. 10 isolated venues, numbered 0..9. Apply union operations [(0, 1), (2, 3), (1, 2), (0, 3), (4, 4)] in this order. How many components remain?

Answer: _____

2. After earlier unions on 10 vertices, two distinct components have sizes 2 and 6; every remaining component is a singleton. Merge those two components. What is the largest component size?

Answer: _____

3. DSU uses union by size and full path compression unless a specific starting state is stated. Consider a find on an uncompressed chain of n vertices. Which is the tightest standard bound among these options?

- A. $O(n^3)$
- B. $O(n)$ for that single find
- C. $O(n!)$
- D. $O(2^n)$

4. An undirected triangle has edge weights [6, 7, 20]. What is its MST cost?

Answer: _____

5. A cable graph has edges (0,1,6), (2,3,8), (3,4,10) and isolated vertex 5. What is the minimum spanning forest cost?

Answer: _____

6. A graph has 11 vertices and 3 connected components. How many edges does every spanning forest have?

Answer: _____

7. Prim has visited {0,1}. Crossing edges are (1,2,9), (0,3,7), (1,3,11). An internal edge (0,1,0) also exists. What is the weight of the next edge chosen?

Answer: _____

8. A flawed Kruskal implementation processes triangle edges in INPUT order: (0,1,8), (1,2,17), (0,2,6), without sorting. It adds an edge when roots differ. By how much does its tree cost exceed the true MST cost?

Answer: _____

9. Directed graph: vertices 0..8, edges [(0, 1), (1, 0), (1, 2), (2, 3), (3, 2)]; other vertices are isolated. How many SCCs are there?

Answer: _____

10. SCCs are $A=\{0,1\}$, $B=\{2\}$, $C=\{3\}$. There are 6 parallel edges $A \rightarrow B$, 7 edges $B \rightarrow C$, one $A \rightarrow C$ and two internal edges in A. How many distinct edges are in the condensation?

Answer: _____

11. Dependencies contain edges [(0, 2), (2, 0), (2, 1), (1, 2)]. Vertices 0,1,2 belong to one SCC. What does replacing that SCC by one node establish?

- A. All original tasks can now be executed in a topological order.
- B. The component graph is acyclic, but the original internal dependency cycle still needs resolution.
- C. The original edges have become undirected.
- D. The SCC must have exactly one vertex.

12. First DFS on G appends vertices to its finish list in this order: [4, 1, 3, 0, 2, 5]. In Kosaraju's second DFS on reversed G, which vertex is considered first?

Answer: _____

13. DAG vertices are 0,1,2,3; edges are [(0, 1), (1, 2)]. How many valid topological orders exist?

Answer: _____

14. DFS reaches a grey vertex through an edge in directed cycle [(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 0)]. Which action is correct?

- A. Report that no topological order exists.
- B. Return the vertices sorted by numeric ID.
- C. Ignore the edge and return reverse postorder.
- D. Treat the edge as an MST candidate.

15. On a DAG, DFS finishes vertices in order [12, 25, 80, 16]. What is the first vertex in reverse postorder?

Answer: _____

16. A topological-ordering algorithm stores n output vertices and n DFS states, excluding the input graph. What is its auxiliary-space bound?

- A. $O(2^n)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \text{ factorial})$

17. DAG edges: (0,1,6), (0,2,9), (1,2,-8), (2,3,4). What is distance(0,3)?

Answer: _____

18. DAG edges 0->1 cost 7, 1->2 cost 3, and 0->2 cost 12. Now change ONLY the weight of 1->2 to -8. What is the new distance from 0 to 2?

Answer: _____

19. For source 47, predecessor[15]=47, predecessor[21]=15, predecessor[16]=21, predecessor[11]=16, predecessor[57]=11, predecessor[12]=57. How many edges are in the reconstructed route to 12?

Answer: _____

20. A unique source-to-target DAG path has 6 edges, each of weight 1,000,000,000. What is its total cost?

Answer: _____